DEMO (fictional, not a real paper): Effect of reaction temperature and time on the yield of aspirin synthesis

A. Demo, B. Example. Department of Imaginary Chemistry. This document was written only to test software.

Abstract

Acetylsalicylic acid (aspirin) was synthesised from salicylic acid and acetic anhydride with sulfuric acid as the catalyst. The reaction temperature was varied from 55 to 90 °C and the reaction time from 5 to 40 min. The highest yield, 85 %, was obtained at 75 °C and 20 min. Lower temperatures gave incomplete conversion, and temperatures above 80 °C reduced the yield.

1. Introduction

Aspirin is made by acetylation of salicylic acid. The reaction is fast, but the isolated yield depends on temperature, time and on how the product is crystallised. This short study reports a simple batch procedure and shows how yield responds to the two main operating conditions.

2. Methods

2.1 Materials

Salicylic acid (99 % purity, 2.0 g) and acetic anhydride (98 % purity, 5.0 mL) were used as received. Concentrated sulfuric acid (5 drops, about 0.25 mL) served as the catalyst. Distilled water (50 mL, ice cold) was used for crystallisation.

2.2 Reaction procedure and equipment

Salicylic acid was weighed on an analytical balance (±0.1 mg) into a 125 mL Erlenmeyer flask. Acetic anhydride and the catalyst were added. The flask was held in a thermostatic water bath at 75 °C (±1 °C) for 20 min with a magnetic stirrer at 300 rpm. The flask was then moved to an ice bath for 15 min, and 50 mL of ice-cold distilled water was added to crystallise the product. The crystals were collected by vacuum filtration on a Büchner funnel and dried in an oven at 50 °C for 24 h.

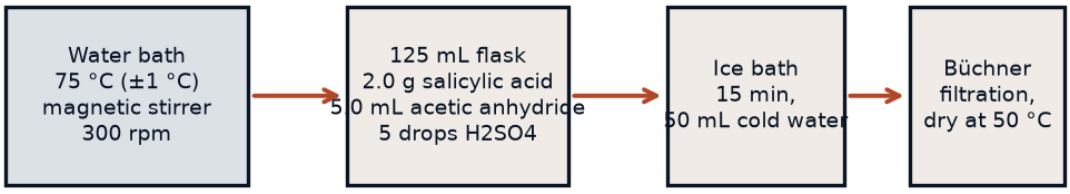


Figure 1. Experimental setup and procedure of the aspirin synthesis (water bath, reaction flask, ice bath, filtration).

2.3 Analysis

The yield was calculated from the dried mass relative to the theoretical mass. The melting point was measured with a capillary melting point apparatus. Each condition was run in triplicate.

3. Results and discussion

3.1 Effect of reaction temperature

The yield (see Fig. 2) at a fixed reaction time of 20 min. The yield rose from 48 % at 55 °C to a maximum of 85 % at 75 °C. Above 80 °C it fell again, to 72 % at 90 °C, because acetic anhydride and aspirin hydrolyse faster at high temperature. At 60 °C the yield was only 62 %.

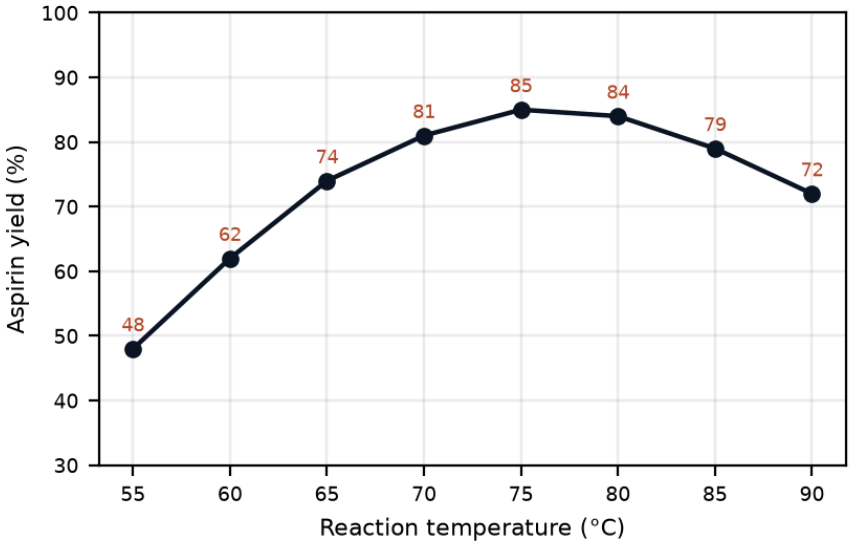


Figure 2. Effect of reaction temperature on aspirin yield (reaction time 20 min, 300 rpm).

3.2 Effect of reaction time

The yield at 75 °C is plotted in Fig. 3. The yield increased from 41 % after 5 min to 85 % after 20 min and then levelled off. Longer times gave no further gain, so 20 min was chosen as the standard time.

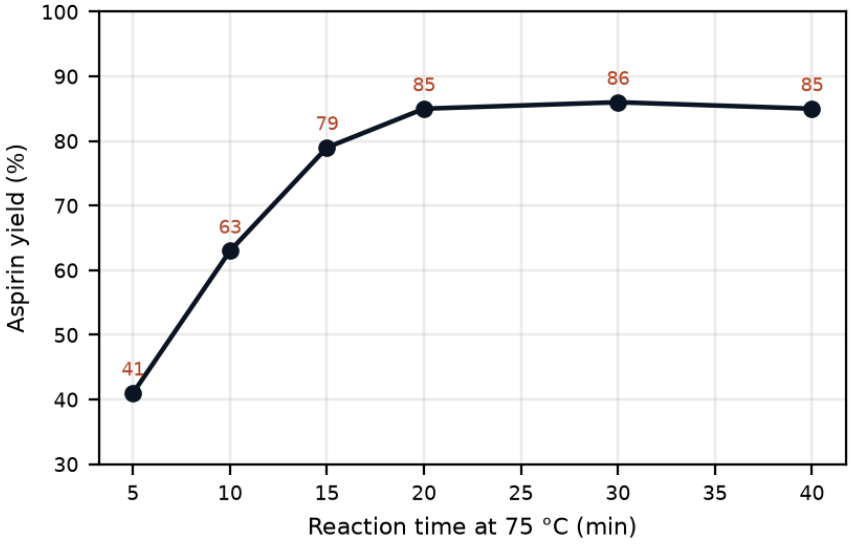


Figure 3. Effect of reaction time on aspirin yield (75 °C, 300 rpm).

3.3 Product quality

Under the standard conditions (75 °C, 20 min, 300 rpm) the product melted at 135 °C and the average yield was 85 %.

4. Conclusions

A temperature of 75 °C and a reaction time of 20 min gave the best aspirin yield (85 %) in this batch procedure.

References

[1] A. Demo, Fictional Journal of Imaginary Chemistry 1 (2026) 1-5. This reference is invented.